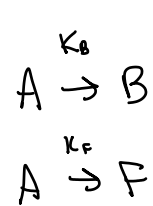


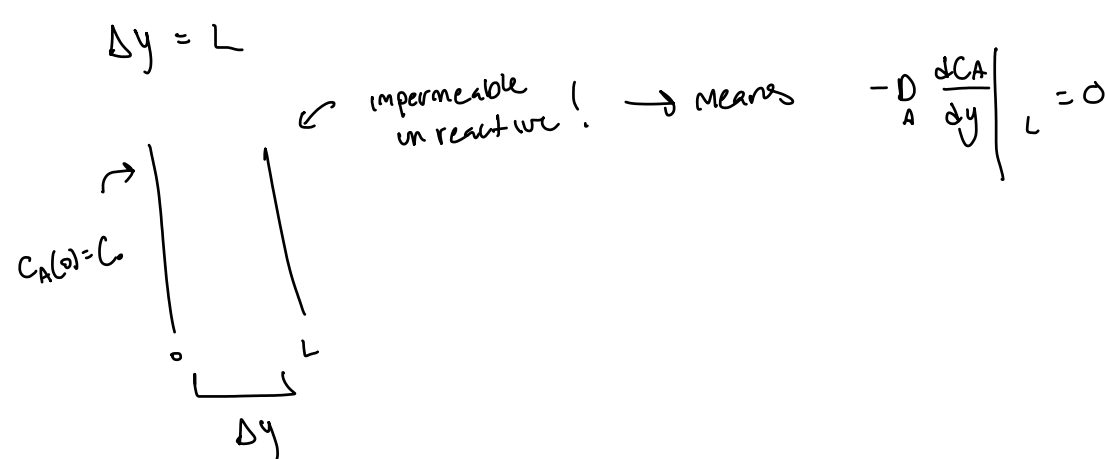
Hw 9

Monday, November 24, 2025

12:00 PM



* homogeneously



$$D_A = D_0 / (1 + \alpha y)$$

$$\frac{d}{dy} \left(D_A \frac{dC_A}{dy} \right) - K_B - K_F = 0$$

$$\rightarrow D_A \frac{dC_A}{dy} = y(K_B + K_F) + C_1$$

$$\frac{dC_A}{dy} = \frac{y(1 + \alpha y)(K_B + K_F)}{D_0} + \frac{(1 + \alpha y) C_1}{D_0}$$

$$\int dy = \int dy$$

$$C_A = \frac{y^2 M}{2 D_0} + \frac{\alpha y^3 M}{3 D_0} + C_1 \left(y + \frac{\alpha y^2}{2} \right) + C_2$$

$$C_A(0) = C_0 = 0 + 0 + 0 = C_2$$

$$C_2 = C_0$$

$$-D_0 \frac{dC_A}{dy} \Big|_L = 0$$

$$\frac{dC_A}{dy} \Big|_L = 0$$

$$0 = \frac{L(1 + \alpha L)(K_B + K_F)}{D_0} + \frac{(1 + \alpha L) C_1}{D_0}$$

$$0 = \frac{(1 + \alpha L)}{D_0} (LM + C_1 D_0)$$

$$\uparrow \neq 0 = 0$$

$$-LM = C_1 D_0$$

$$\frac{-LM}{D_0} = C_1$$

Overall rate of consumption

$$N_A(0) = -D_A(0) \frac{dC_A}{dy} \Big|_0 =$$

$$\cancel{\left((0)(K_B + K_F) + (1 + \alpha \cdot 0) C_1 \right)}$$

$$\frac{D_0}{(1 + \alpha \cdot 0)} \left(C_1 = \frac{-LM}{D_0} \right)$$

$$= -LM \text{ overall consumption}$$

Q₂)

$$Q = m_n c_p (T_{in} - T_{out}) = 1.8 \times 10^4 \text{ W}$$

$$= m_c c_p (T_{out} - T_{in})$$

↪
?

$$\frac{Q}{m_c c_p} + T_{in} = T_{out} = 82.5^\circ \text{C}$$

Counter-current

$$\Delta T_m = \frac{(T_{in} - T_{out}) - (T_{out} - T_{in})}{\ln \left(\frac{T_{in} - T_{out}}{T_{out} - T_{in}} \right)} = 83.7^\circ \text{C}$$

$$Q = A U \Delta T_m$$

$$A = \frac{Q}{U \Delta T_m} = 0.84 \text{ m}^2$$

Parallel

$$\Delta T_m = \frac{(T_{in} - T_{in}) - (T_{out} - T_{out})}{\ln \left(\frac{T_{in} - T_{in}}{T_{out} - T_{out}} \right)} = 49.2^\circ \text{C}$$

$$A = 0.92 \text{ m}^2$$